Alexandria University
Faculty of Engineering
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Assignment #1
Multi-class Diabetes Risk Prediction

Module: K-Nearest Neighbours Classifier

Course:	CSE – Pattern Recognition
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1 Introduction

This report presents the K-Nearest Neighbours (KNN) component of Assignment #1, which addresses multi-class diabetes risk prediction using the *Diabetes Health Indicators Dataset* (BRFSS 2015). The dataset contains 21 patient attributes (e.g. BMI, age, high blood pressure, physical activity) and a three-class target variable: **0** (No Diabetes), **1** (Prediabetes), and **2** (Diabetes). A central challenge is severe class imbalance: the Prediabetes class is substantially smaller than the other two.

The goals of this module are to:

(i) tune a KNN classifier on the validation set across multiple values of k and two distance metrics;

(ii) train a baseline model on the original imbalanced data and a second model after applying Synthetic Minority Oversampling (SMOTE);

(iii) evaluate both models on the held-out test set using accuracy, micro-, macro-, and weighted F1-scores; and

(iv) analyse the confusion matrices and the impact of SMOTE on minority-class recall.

2 Data Preparation

The full data pipeline is implemented in `data.py`. The key steps are summarised below.

2.1 Train / Validation / Test Split

The dataset is partitioned with stratified sampling to preserve the original class distribution in each subset:

Split	Fraction	Approx. samples
Training	70%	177,580
Validation	10%	25,369
Test	20%	50,736

A fixed random seed of `42` is used throughout to guarantee reproducibility, as required by the assignment.

2.2 Feature Preprocessing

All features are standardised with `StandardScaler` (zero mean, unit variance) fitted exclusively on the training partition. This step is critical for KNN because the algorithm is distance-based; without scaling, high-variance features such as BMI would dominate the distance computation.

2.3 Feature Selection

To reduce dimensionality and remove uninformative features, `SelectKBest` with `mutual_info_classif` is used to retain the top $K = 15$ features out of the original 21. The selector is fitted on the training set only to prevent data leakage, and the learned mask is then applied identically to the validation and test sets.

2.4 Class Imbalance

The original class distribution is heavily skewed towards No Diabetes. The Prediabetes class (class 1) accounts for less than 2% of all samples, making this a severe imbalance problem. For the *balanced* variant of KNN, Synthetic Minority Oversampling Technique (SMOTE) is applied to the training set after feature selection, generating synthetic interpolated samples for the minority classes until all three classes are equally represented.

3 KNN Implementation and Hyperparameter Tuning

3.1 Algorithm Overview

The K-Nearest Neighbours classifier is a non-parametric, instance-based learning algorithm. Given a query point $\mathbf{x}$, it identifies the k training samples closest in feature space and assigns the majority class label:

$$\hat{y}(\mathbf{x}) = \arg\max_{c} \sum_{i \in \mathcal{N}_k(\mathbf{x})} \mathbf{1}[y_i = c],$$

where $\mathcal{N}_k(\mathbf{x})$ is the set of k nearest neighbours under a chosen distance metric.

Two distance metrics were evaluated:

- **Euclidean** (L_2): $d(\mathbf{x}, \mathbf{z}) = \sqrt{\sum_j (x_j - z_j)^2}$
- **Manhattan** (L_1): $d(\mathbf{x}, \mathbf{z}) = \sum_j |x_j - z_j|$

3.2 Hyperparameter Search

The search grid spans $k \in \{3, 5, 11, 21\}$ and both distance metrics. For each combination, a KNN model is trained on the appropriate training set (imbalanced for the baseline, SMOTE-resampled for the balanced variant) and evaluated on the validation set using **macro F1-score** – the most informative metric for an imbalanced dataset (see Section 4.2).

Baseline search (imbalanced data).

Method	k	Metric	Val. Macro F1
none	3	euclidean	0.3935
none	3	manhattan	**0.3955**
none	5	euclidean	0.3904
none	5	manhattan	0.3917
none	11	euclidean	0.3784
none	11	manhattan	0.3816
none	21	euclidean	0.3763
none	21	manhattan	0.3728

Table 1: Baseline (imbalanced) KNN hyperparameter grid on validation set. Best configuration highlighted in bold.

SMOTE search (balanced data).

Method	k	Metric	Val. Macro F1
smote	3	euclidean	0.3998
smote	3	manhattan	0.4036
smote	5	euclidean	0.3971
smote	5	manhattan	**0.4065**
smote	11	euclidean	0.3894
smote	11	manhattan	0.4041
smote	21	euclidean	0.3837
smote	21	manhattan	0.4007

Table 2: SMOTE-balanced KNN hyperparameter grid on validation set. Best configuration highlighted in bold.

Selected configurations.

- **Baseline:** $k = 3$, Manhattan distance (val. macro F1 = 0.3955).
- **Balanced (SMOTE):** $k = 5$, Manhattan distance (val. macro F1 = 0.4065).

Both searches consistently favour Manhattan distance over Euclidean and smaller values of k, suggesting that the decision boundaries in this 15-dimensional scaled feature space benefit from finer local structure. Larger k values lead to over-smoothing, blurring the boundaries around the minority Prediabetes class.

4 Evaluation

4.1 Metrics Reported

Four metrics are computed on the held-out test set:

- **Accuracy:** fraction of correctly classified samples.
- **Micro F1:** globally aggregated precision and recall across all instances; equivalent to accuracy in a single-label setting.
- **Macro F1:** unweighted mean of per-class F1-scores; each class contributes equally regardless of size.
- **Weighted F1:** mean of per-class F1-scores weighted by class support; penalises poor performance on large classes more than on small ones.

4.2 Which Metric Is Most Meaningful?

For this severely imbalanced three-class problem, **macro F1-score** is the most informative metric. Accuracy and micro F1 are dominated by the majority class (No Diabetes, $\approx 84\%$ of samples) and can appear deceptively high even when the model entirely ignores Prediabetes. Weighted F1 down-weights the minority class contributions proportionally to support, again concealing poor minority-class performance. Macro F1, by treating every class equally, directly penalises the model when it fails to recognise Prediabetes or Diabetes cases – the clinically critical misclassifications.

4.3 Test Set Results

Model	Accuracy	Micro F1	Macro F1	Weighted F1
KNN Baseline (Imbalanced)	0.8215	0.8215	0.3992	0.7989
KNN Balanced (SMOTE)	0.7080	0.7080	0.4101	0.7459

Table 3: KNN test-set performance summary.

Per-class breakdown.

Class	Precision	Recall	F1	Support
KNN Baseline (Imbalanced)				
No Diabetes	0.87	0.94	0.90	42,741
Prediabetes	0.05	0.00	0.01	926
Diabetes	0.37	0.24	0.29	7,069
KNN Balanced (SMOTE)				
No Diabetes	0.91	0.76	0.82	42,741
Prediabetes	0.03	0.08	0.04	926
Diabetes	0.29	0.50	0.37	7,069

Table 4: Per-class classification report for both KNN variants.

5 Confusion Matrices

Figure 1 shows the row-normalised confusion matrices for both KNN variants on the test set. Raw sample counts are overlaid in grey.

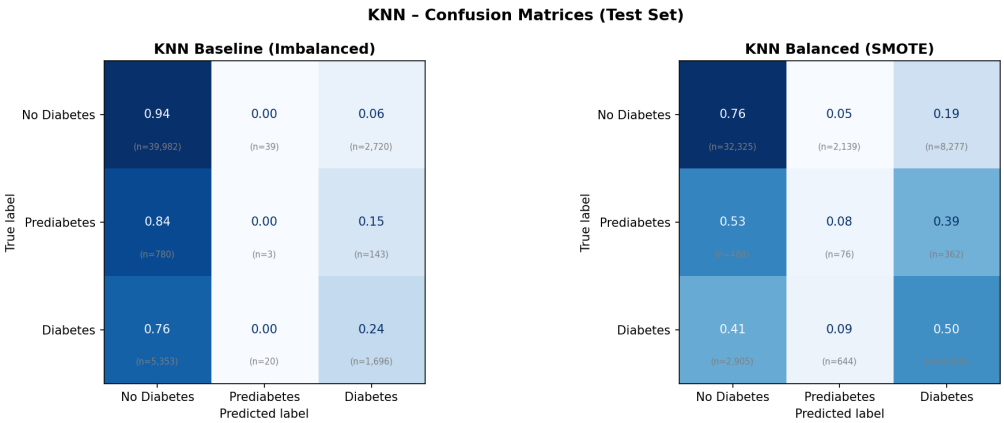

Figure 1: Row-normalised confusion matrices for KNN Baseline (left) and KNN Balanced – SMOTE (right) on the test set. Cell values are row-normalised probabilities; grey annotations give raw counts.

6 Analysis and Discussion

6.1 Impact of SMOTE on Performance

Comparing the two variants reveals a clear trade-off between overall accuracy and minority-class recall:

- **Accuracy / Micro F1** drops from 0.8215 to 0.7080 after SMOTE. This is expected: the model sacrifices some correct predictions for No Diabetes in order to improve recall for the minority classes.
- **Macro F1** *improves* slightly from 0.3992 to 0.4101. Although the gain is modest, it reflects a real improvement in the model's ability to recognise Prediabetes and Diabetes cases.
- **Diabetes recall** improves substantially from 0.24 to 0.50, meaning the SMOTE model correctly identifies twice as many diabetic patients. This is clinically significant.
- **Prediabetes recall** improves from a near-zero 0.00 to 0.08. The absolute level is still very low due to the extreme rarity of this class ($< 2\%$ of samples), but the direction of change is correct.
- **No Diabetes recall** decreases from 0.94 to 0.76. A fraction of the majority class is now misclassified as Diabetes or Prediabetes – the cost of improved minority sensitivity.

6.2 Misclassification Patterns

From Figure 1, several patterns are evident:

Baseline model. The Prediabetes row is almost entirely classified as No Diabetes (0.84) or Diabetes (0.15), with only 3 out of 926 Prediabetes instances correctly identified. The model has effectively learnt to ignore Prediabetes due to its negligible share of the training data. Diabetes recall is also modest at 0.24, with 76% of diabetic patients misclassified as No Diabetes.

Balanced (SMOTE) model. Prediabetes confusion is redistributed: 53% are misclassified as No Diabetes and 39% as Diabetes. The diagonal entry rises from 0.00 to 0.08, indicating the model has learned at least some Prediabetes signal. Diabetes recall jumps to 0.50, but a large proportion (0.41) of diabetic patients are still predicted as No Diabetes.

Clinical interpretation. The systematic confusion between Prediabetes and No Diabetes arises partly because prediabetes is a *transitional* state; its physiological markers (e.g. slightly elevated BMI or borderline blood pressure) overlap substantially with the healthy class. Mathematically, KNN's local neighbourhood in this 15-dimensional feature space tends to be dominated by No Diabetes neighbours due to its overwhelming prevalence, even after SMOTE resampling of the training set.

6.3 Strengths and Limitations of KNN

- **Strengths:** no parametric assumptions; directly captures non-linear decision boundaries; easy to interpret (k nearest examples).

- **Limitations:** inference time scales with training set size ($O(n)$ per query), making it expensive on the full BRFSS dataset; susceptible to the curse of dimensionality; does not naturally model posterior class probabilities; highly sensitive to irrelevant features even after feature selection.

7 Conclusion

The KNN classifier was tuned over a grid of $k \in \{3, 5, 11, 21\}$ and two distance metrics using macro F1-score on the validation set. The best configuration for the baseline was $k = 3$, Manhattan distance, and for the SMOTE-balanced variant $k = 5$, Manhattan distance.

SMOTE modestly improved macro F1 from 0.3992 to 0.4101 and substantially improved Diabetes recall ($0.24 \rightarrow 0.50$), demonstrating its value for surfacing minority-class patterns. However, Prediabetes recall remains very low in both variants due to the extreme class imbalance and the high degree of feature-space overlap with the No Diabetes class.

These results motivate the subsequent evaluation of more expressive models (Softmax Regression and Feedforward Neural Network) which may better separate the overlapping class regions through learned non-linear transformations.